Assignment 10 - Assignment

Topic: K-Means Clustering

Dataset: Assignment_10b.csv

CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40
6	Female	22	17	76
7	Female	35	18	6
8	Female	23	18	94
9	Male	64	19	3
10	Female	30	19	72
11	Male	67	19	14
12	Female	35	19	99
13	Female	58	20	15
14	Female	24	20	77
15	Male	37	20	13
16	Male	22	20	79
17	Female	35	21	35
18	Male	20	21	66
19	Male	52	23	29
20	Female	35	23	98
21	Male	35	24	35
22	Male	25	24	73
23	Female	46	25	5
24	Male	31	25	73
25	Female	54	28	14
26	Male	29	28	82
27	Female	45	28	32
28	Male	35	28	61
29	Female	40	29	31
30	Female	23	29	87
31	Male	60	30	4
32	Female	21	30	73
33	Male	53	33	4
34	Male	18	33	92
35	Female	49	33	14
36	Female	21	33	81
37	Female	42	34	17
38	Female	30	34	73
39	Female	36	37	26
40	Female	20	37	75
41	Female	65	38	35
42	Male	24	38	92
43	Male	48	39	36
44	Female	31	39	61
45	Female	49	39	28
46	Female	24	39	65
47	Female	50	40	55
48	Female	27	40	47
49	Female	29	40	42
50	Female	31	40	42
51	Female	49	42	52
52	Male	33	42	60
53	Female	31	43	54
54	Male	59	43	60
55	Female	50	43	45
56	Male	47	43	41
57	Female	51	44	50
58	Male	69	44	46
59	Female	27	46	51
60	Male	53	46	46
61	Male	70	46	56

CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
62	Male	19	46	55
63	Female	67	47	52
64	Female	54	47	59
65	Male	63	48	51
66	Male	18	48	59
67	Female	43	48	50
68	Female	68	48	48
69	Male	19	48	59
70	Female	32	48	47
71	Male	70	49	55
72	Female	47	49	42
73	Female	60	50	49
74	Female	60	50	56
75	Male	59	54	47
76	Male	26	54	54
77	Female	45	54	53
78	Male	40	54	48
79	Female	23	54	52
80	Female	49	54	42
81	Male	57	54	51
82	Male	38	54	55
83	Male	67	54	41
84	Female	46	54	44
85	Female	21	54	57
86	Male	48	54	46
87	Female	55	57	58
88	Female	22	57	55
89	Female	34	58	60
90	Female	50	58	46
91	Female	68	59	55
92	Male	18	59	41
93	Male	48	60	49
94	Female	40	60	40
95	Female	32	60	42
96	Male	24	60	52
97	Female	47	60	47
98	Female	27	60	50
99	Male	48	61	42
100	Male	20	61	49
101	Female	23	62	41
102	Female	49	62	48
103	Male	67	62	59
104	Male	26	62	55
105	Male	49	62	56
106	Female	21	62	42
107	Female	66	63	50
108	Male	54	63	46
109	Male	68	63	43
110	Male	66	63	48
111	Male	65	63	52
112	Female	19	63	54
113	Female	38	64	42
114	Male	19	64	46
115	Female	18	65	48
116	Female	19	65	50
117	Female	63	65	43
118	Female	49	65	59
119	Female	51	67	43
120	Female	50	67	57
121	Male	27	67	56
122	Female	38	67	40
123	Female	40	69	58
124	Male	39	69	91
125	Female	23	70	29
126	Female	31	70	77
127	Male	43	71	35
128	Male	40	71	95
129	Male	59	71	11

CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
130	Male	38	71	75
131	Male	47	71	9
132	Male	39	71	75
133	Female	25	72	34
134	Female	31	72	71
135	Male	20	73	5
136	Female	29	73	88
137	Female	44	73	7
138	Male	32	73	73
139	Male	19	74	10
140	Female	35	74	72
141	Female	57	75	5
142	Male	32	75	93
143	Female	28	76	40
144	Female	32	76	87
145	Male	25	77	12
146	Male	28	77	97
147	Male	48	77	36
148	Female	32	77	74
149	Female	34	78	22
150	Male	34	78	90
151	Male	43	78	17
152	Male	39	78	88
153	Female	44	78	20
154	Female	38	78	76
155	Female	47	78	16
156	Female	27	78	89
157	Male	37	78	1
158	Female	30	78	78
159	Male	34	78	1
160	Female	30	78	73
161	Female	56	79	35
162	Female	29	79	83
163	Male	19	81	5
164	Female	31	81	93
165	Male	50	85	26
166	Female	36	85	75
167	Male	42	86	20
168	Female	33	86	95
169	Female	36	87	27
170	Male	32	87	63
171	Male	40	87	13
172	Male	28	87	75
173	Male	36	87	10
174	Male	36	87	92
175	Female	52	88	13
176	Female	30	88	86
177	Male	58	88	15
178	Male	27	88	69
179	Male	59	93	14
180	Male	35	93	90
181	Female	37	97	32
182	Female	32	97	86
183	Male	46	98	15
184	Female	29	98	88
185	Female	41	99	39
186	Male	30	99	97
187	Female	54	101	24
188	Male	28	101	68
189	Female	41	103	17
190	Female	36	103	85
191	Female	34	103	23
192	Female	32	103	69
193	Male	33	113	8
194	Female	38	113	91
195	Female	47	120	16
196	Female	35	120	79
197	Female	45	126	28

CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
198	Male	32	126	74
199	Male	32	137	18
200	Male	30	137	83

Question 1

Load the Mall Customers dataset from the CSV file. Display the first 10 rows and identify the input features and target variable.

Dataset: kaggle.com

[1]:
```
import pandas as pd

# Load the Mall Customers dataset
df = pd.read_csv('Assignment_10b.csv')

# Display first 10 rows
print(df.head(10).to_string())

# Identify features
print('\nShape:', df.shape)
print('\nColumns:', list(df.columns))
print('\nFeatures available for clustering:')
print('  - Age')
print('  - Annual Income (k$)')
print('  - Spending Score (1-100)')
print('\nNote: CustomerID is just an identifier and Gender is categorical.
 ↪')
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
5           6  Female   22                  17                      76
6           7  Female   35                  18                       6
7           8  Female   23                  18                      94
8           9    Male   64                  19                       3
9          10  Female   30                  19                      72

Shape: (200, 5)

Columns: ['CustomerID', 'Gender', 'Age', 'Annual Income (k$)', 'Spending␣
 ↪Score
(1-100)']

Features available for clustering:
  - Age
  - Annual Income (k$)
```

```
  - Spending Score (1-100)

Note: CustomerID is just an identifier and Gender is categorical.
```

Question 2

Select two features — Annual Income (k$) and Spending Score (1-100) — for clustering.

[2]:
```python
print('----- Selected Features -----')

# Select two features for clustering
X = df[['Annual Income (k$)', 'Spending Score (1-100)']].values

print('\nFeature 1: Annual Income (k$)')
print('Feature 2: Spending Score (1-100)')
print('\nThese two features are chosen because they directly reflect')
print('customer purchasing power and spending behaviour.')
print('\nShape of selected feature matrix:', X.shape)
print()
print(df[['Annual Income (k$)', 'Spending Score (1-100)']].head(10).
 ↪to_string())
```

```
----- Selected Features -----

Feature 1: Annual Income (k$)
Feature 2: Spending Score (1-100)

These two features are chosen because they directly reflect
customer purchasing power and spending behaviour.

Shape of selected feature matrix: (200, 2)

   Annual Income (k$)  Spending Score (1-100)
0                  15                      39
1                  15                      81
2                  16                       6
3                  16                      77
4                  17                      40
5                  17                      76
6                  18                       6
7                  18                      94
8                  19                       3
9                  19                      72
```

Question 3

Apply K-Means clustering with K = 5 on the selected features. Print the cluster labels and cluster centroids.

[3]:
```python
from sklearn.cluster import KMeans

print('----- K-Means Clustering (K=5) -----')
```

```
# Apply K-Means with K=5
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(X)
centroids = kmeans.cluster_centers_

print('\nK-Means Model Fitted Successfully.')
print('\nCluster Labels (first 20):', cluster_labels[:20].tolist())
print('\nCluster Centroids:')
for i, c in enumerate(centroids):
    print(f'  Cluster {i}: Annual Income = {c[0]:.2f} k$, Spending Score␣
 ↪= {c[1]:.2f}')
```

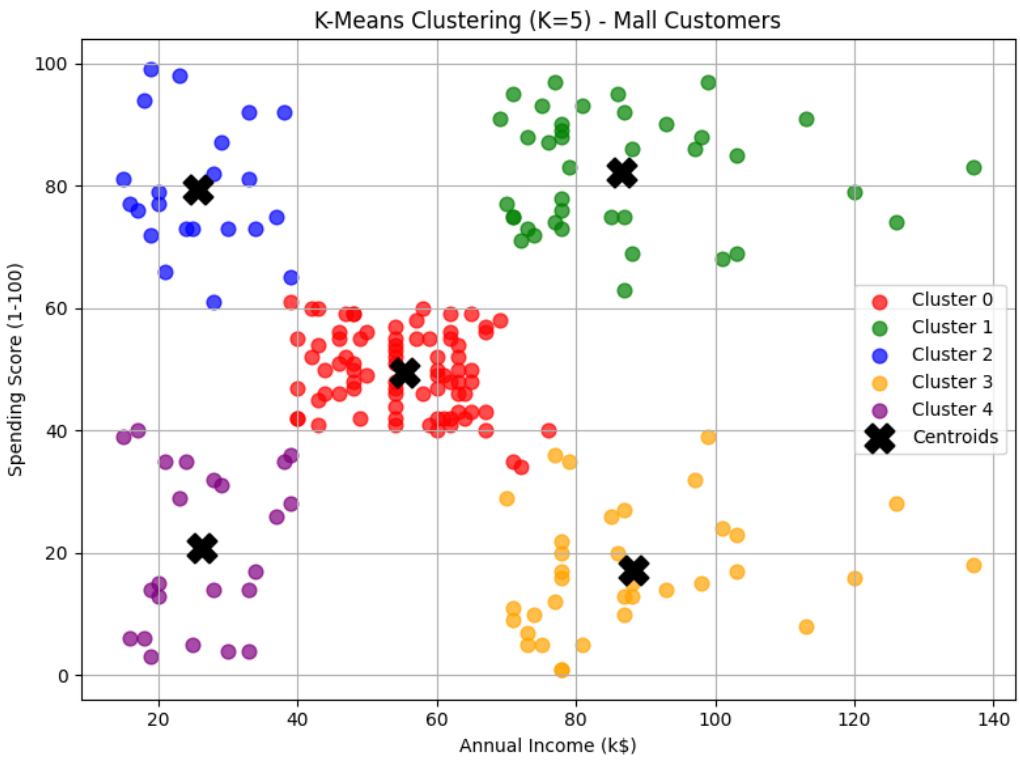
```
----- K-Means Clustering (K=5) -----

K-Means Model Fitted Successfully.

Cluster Labels (first 20): [4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2,␣
 ↪4,
2, 4, 2]

Cluster Centroids:
  Cluster 0: Annual Income = 55.30 k$, Spending Score = 49.52
  Cluster 1: Annual Income = 86.54 k$, Spending Score = 82.13
  Cluster 2: Annual Income = 25.73 k$, Spending Score = 79.36
  Cluster 3: Annual Income = 88.20 k$, Spending Score = 17.11
  Cluster 4: Annual Income = 26.30 k$, Spending Score = 20.91
```

Question 4

Plot a scatter plot showing the K-Means clusters and mark the cluster centroids.

[4]:
```
import matplotlib.pyplot as plt

colors = ['red', 'green', 'blue', 'orange', 'purple']

plt.figure(figsize=(8, 6))
for c in range(5):
    mask = cluster_labels == c
    plt.scatter(X[mask, 0], X[mask, 1], c=colors[c],
                label=f'Cluster {c}', alpha=0.7, s=60)

# Mark centroids
plt.scatter(centroids[:, 0], centroids[:, 1], c='black', marker='X',
            s=250, linewidths=1.5, label='Centroids', zorder=5)

plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('K-Means Clustering (K=5) - Mall Customers')
plt.legend()
plt.grid(True)
```

```
plt.tight_layout()
plt.show()
```

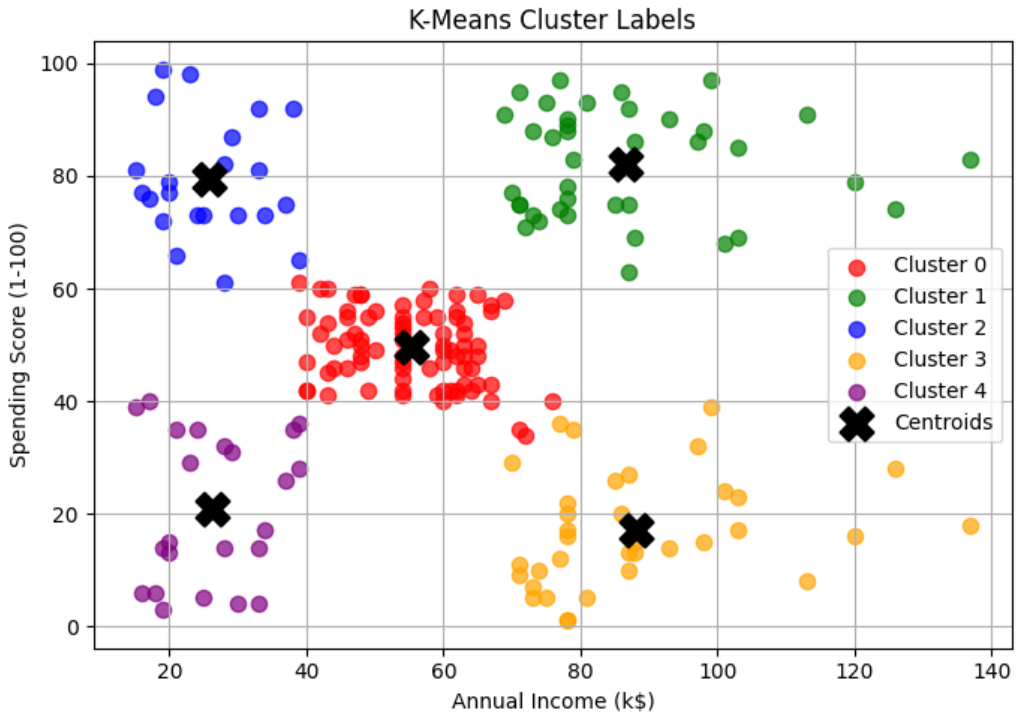


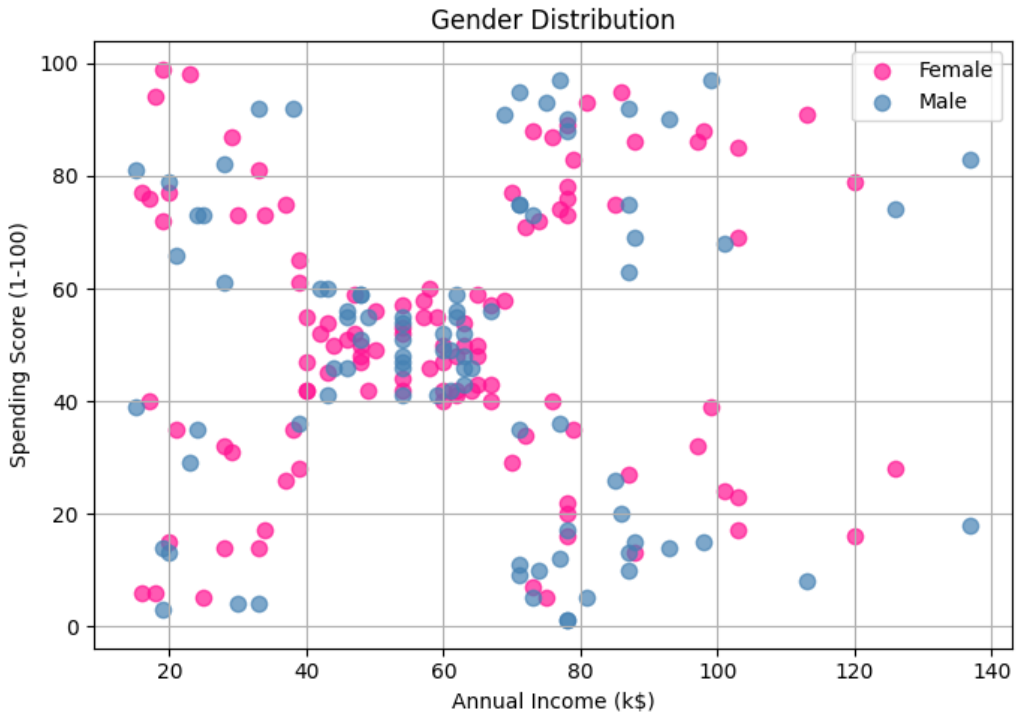
Question 5

Compare the cluster labels from K-Means with the Gender attribute. Plot both side by side and summarise each cluster.

[5]:
```
from sklearn.preprocessing import LabelEncoder

# Encode Gender for reference comparison
le = LabelEncoder()
gender_labels = le.fit_transform(df['Gender'])   # Female=0, Male=1
gender_colors = ['deeppink', 'steelblue']
gender_names  = le.classes_

# --- Plot 1: K-Means Clusters ---
fig1, ax1 = plt.subplots(figsize=(7, 5))
for c in range(5):
    mask = cluster_labels == c
    ax1.scatter(X[mask, 0], X[mask, 1], c=colors[c],
                label=f'Cluster {c}', alpha=0.7, s=55)
ax1.scatter(centroids[:, 0], centroids[:, 1], c='black', marker='X',
            s=250, linewidths=1.5, label='Centroids', zorder=5)
ax1.set_xlabel('Annual Income (k$)')
ax1.set_ylabel('Spending Score (1-100)')
```

```python
ax1.set_title('K-Means Cluster Labels')
ax1.legend()
ax1.grid(True)
plt.tight_layout()
plt.show()

# --- Plot 2: Gender Distribution ---
fig2, ax2 = plt.subplots(figsize=(7, 5))
for g in range(2):
    mask = gender_labels == g
    ax2.scatter(X[mask, 0], X[mask, 1],
                c=gender_colors[g], label=gender_names[g],
                alpha=0.7, s=55)
ax2.set_xlabel('Annual Income (k$)')
ax2.set_ylabel('Spending Score (1-100)')
ax2.set_title('Gender Distribution')
ax2.legend()
ax2.grid(True)
plt.tight_layout()
plt.show()

# Cluster summary table
comp_df = df[['CustomerID', 'Gender', 'Age',
              'Annual Income (k$)', 'Spending Score (1-100)']].copy()
comp_df['Cluster'] = cluster_labels
summary = comp_df.groupby('Cluster')[['Age', 'Annual Income (k$)',
                                      'Spending Score (1-100)']].mean().
 ↪round(2)
summary.columns = ['Avg Age', 'Avg Income (k$)', 'Avg Spending Score']
print('----- Comparison: K-Means Clusters vs Gender -----')
print('\nCluster Summary (mean values per cluster):')
print(summary.to_string())
print('\nGender distribution per cluster:')
print(comp_df.groupby(['Cluster', 'Gender']).size().to_string())
print('\nNote: Since this dataset has no definitive ground-truth cluster␣
 ↪labels,')
print('      Gender is used as a reference attribute for comparison.')
```






```
----- Comparison: K-Means Clusters vs Gender -----

Cluster Summary (mean values per cluster):
          Avg Age  Avg Income (k$)  Avg Spending Score
```

```
Cluster
0          42.72            55.30              49.52
1          32.69            86.54              82.13
2          25.27            25.73              79.36
3          41.11            88.20              17.11
4          45.22            26.30              20.91

Gender distribution per cluster:
Cluster  Gender
0        Female    48
         Male      33
1        Female    21
         Male      18
2        Female    13
         Male       9
3        Female    16
         Male      19
4        Female    14
         Male       9

Note: Since this dataset has no definitive ground-truth cluster labels,
      Gender is used as a reference attribute for comparison.
```

Question 6

Calculate and print the Adjusted Rand Index (ARI) to evaluate clustering performance.

[6]:
```python
from sklearn.metrics import adjusted_rand_score

print('----- Adjusted Rand Index (ARI) -----')

# ARI between K-Means clusters and Gender labels
ari = adjusted_rand_score(gender_labels, cluster_labels)

print(f'\nARI Score (K-Means clusters vs Gender): {ari:.4f}')
print('\nInterpretation:')
print('  ARI = 1.0  → Perfect match with reference labels')
print('  ARI = 0.0  → No better than random assignment')
print('  ARI < 0.0  → Worse than random assignment')
print(f'\nARI = {ari:.4f} suggests that the K-Means clusters are')
print('largely independent of gender - which is expected, since')
print('clustering is driven by income and spending behaviour, not gender.
 ↪')
print('\nA better evaluation for this dataset is the Within-Cluster Sum␣
 ↪of')
print('Squares (Inertia), which K-Means minimises internally:')
print(f'  Inertia (WCSS): {kmeans.inertia_:.2f}')
```

```
----- Adjusted Rand Index (ARI) -----

ARI Score (K-Means clusters vs Gender): -0.0006
```

```
Interpretation:
  ARI = 1.0  → Perfect match with reference labels
  ARI = 0.0  → No better than random assignment
  ARI < 0.0  → Worse than random assignment

ARI = -0.0006 suggests that the K-Means clusters are
largely independent of gender - which is expected, since
clustering is driven by income and spending behaviour, not gender.

A better evaluation for this dataset is the Within-Cluster Sum of
Squares (Inertia), which K-Means minimises internally:
  Inertia (WCSS): 44448.46
```